

Python Semester Project

Real-Time Data Plotter

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Animated Line Plot

Constructed and animated figure using the matplotlib package, pyplot and animation modules.

```
def animate(frame, xs, ya, yb):
    timestamp, data = get_data()
    xs.append(timestamp)

    #limit length of time (x) list
    xs=xs[-x_len:]
    ax.set_xlim(xs[0],xs[-1])

    #limit length of rate (y) lists
    ya.append(data.ask)
    yb.append(data.bid)
    ya=ya[-x_len:]
    yb=yb[-x_len:]

    #adjust y limits for new data point
    y_limits = ax.get_ylim()
    up_lim = y_limits[1]
    low_lim = y_limits[0]

    if(data.bid < low_lim):
        low_lim = data.bid-1
    if(data.ask > up_lim):
        up_lim = data.ask+1

    ax.set_ylim(low_lim, up_lim)

    ax.set_xticks(xs)
    plt.xticks(rotation=45, ha='right')

    #update line
    line1.set_data(xs,ya)
    line2.set_data(xs,yb)
```

Real-Time Data

Pulled latest price data from
n.exchange for bitcoin to
litecoin rates.

Extracted the bidding and
asking prices for plotting.

```
ex_response = [  
    {  
        "created_on": "2026-05-04T01:29:42.558250Z",  
        "unix_time": "1777858182.0",  
        "expiration_time_unix": "1777858213.0",  
        "ticker": {  
            "ask": "1426.53352354",  
            "bid": "1424.50142450"  
        },  
        "market": {  
            "name": "n.exchange",  
            "code": "nex"  
        },  
        "rate": "1425.51747401961382",  
        "allowed_historic_price_variance": true,  
        "allowed_historic_price_age": true  
    }  
]
```